

The Fundamental Theorem of Resolution Geometry

A Procedural Derivation of the Synthesis

as Necessity and Sufficiency

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Abstract

The synthesis of resolution geometry was previously stated one-way: the elimination of the null sector *generates* the local and global structure. This paper proves the two-way statement—necessity, sufficiency, and uniqueness—by derivation rather than numerical verification, assembling three established theorems and checking their mutual consistency. The engine is the fundamental theorem of submanifolds, itself an equivalence: a first fundamental form, a second fundamental form, and a normal connection are realized by an isometric immersion, unique up to ambient isometry, *if and only if* they satisfy the Gauss, Codazzi, and Ricci equations. We combine it with the Schur completion theorem (an equivalence for first-order covariance admissibility) and the Luna slice theorem (classification on every stratum), linking the statistical and the embedding geometry by the Mahalanobis identification $g = \Sigma^{-1}$, which makes the induced metric equal to the Fisher metric. The result is a fundamental theorem of resolution geometry: a geometry exists and is unique up to gauge iff its first-order residue satisfies positive-definiteness and the Schur floor and its second-order residue satisfies Gauss–Codazzi–Ricci with respect to the covariance-induced metric. We then perform the consistency check the derivation demands: the first- and second-order conditions constrain distinct data, are linked by the Gauss equation, and are not over-determined, so no contradiction arises; the only degeneracy is the covariance-support boundary where the Mahalanobis metric fails. Finally we make the globalization rigorous without topological hand-waving: because local immersions are unique up to ambient isometry, patch overlaps differ by ambient isometries and therefore form a genuine $\text{Aut}(\mathcal{G})$ -valued cocycle *by the theorem*, and the obstruction to a global immersion is precisely the monodromy representation $\pi_1 \rightarrow \text{Aut}(\mathcal{G})$ —the character variety, now identified as the standard monodromy obstruction rather than an ad hoc object. The honest verdict: the synthesis is a genuine equivalence *locally* (necessity, sufficiency, uniqueness all hold) and an equivalence *globally modulo monodromy*; the computation of the monodromy (the character variety) remains the one open problem, and it is a problem of mathematics at large, not of the derivation.

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1 Aim: from generation to equivalence

The prior synthesis showed that the null-sector elimination *generates* the structure: geometry $\Rightarrow$ residues $\Rightarrow$ holonomy. That is one implication. A credible spine requires the converse and the uniqueness: that data satisfying the right conditions *is* a resolution geometry, and a unique one up to gauge. This paper supplies the missing directions by assembling established equivalences, and it checks that the assembled conditions are mutually consistent. No step relies on numerical verification.

We fix notation. Let $Y : \Theta \rightarrow (O, g)$ be a smooth observation map into an observation space with metric g , $J = dY$ its differential, $R = \text{Im } J$ the resolved sector, $N = R^\perp_g = \text{coker } J$ the null sector, and Σ the observation covariance with blocks (Σ_R, C, Σ_N) in the $R \oplus N$ split.

2 The Mahalanobis identification

The statistical and embedding geometries are unified by one choice of metric.

Assumption 2.1 (Mahalanobis metric). The observation metric is the inverse covariance, $g = \Sigma^{-1}$, on the support of Σ (where $\Sigma \succ 0$).

Lemma 2.2 (Induced metric equals Fisher metric). *Under Assumption 2.1, the metric induced on Θ by the immersion Y , namely $Y^*g = J^\top g J$, equals the Fisher metric $J^\top \Sigma^{-1} J$. Hence the submanifold geometry of Y and the information geometry of the model coincide.*

Proof. $Y^*g = J^\top g J = J^\top \Sigma^{-1} J$ by Assumption 2.1, which is the Fisher metric [7]. The resolved/null split $R \oplus N$ is the g -orthogonal, hence Σ^{-1} -orthogonal, split. $\square$

Remark 2.3. Assumption 2.1 is the statistically canonical choice: it is the metric in which distances are Mahalanobis distances. It fails only on the covariance kernel, i.e. at the support boundary $R \not\subseteq \text{range } \Sigma$, which is exactly the finite-information boundary already isolated in the corpus. We work on the support throughout and return to the boundary in §6.

3 First-order layer: the Schur completion equivalence

Lemma 3.1 (Reduced isomorphism and index). *J restricts to a linear isomorphism $(\ker J)^\perp \xrightarrow{\sim} R$, and $\text{ind } J = \dim \ker J - \dim N = \dim \Theta - \dim O$ is independent of J .*

Proof. Rank–nullity: J is injective on $(\ker J)^\perp$ with image $R = \text{Im } J$; $\dim \ker J = \dim \Theta - \text{rank } J$ and $\dim N = \dim O - \text{rank } J$, whose difference is $\dim \Theta - \dim O$ [8]. $\square$

Theorem 3.2 (Schur completion: first-order admissibility, iff). *Given $\Sigma_R \succ 0$ and C , a symmetric Σ_N completes Σ to a positive semidefinite (resp. definite) matrix if and only if*

$$\Sigma_N \succeq C^\top \Sigma_R^{-1} C \quad (\text{resp. } \succ).$$

The set of admissible Σ_N is a closed convex cone; the completion realizing a prescribed Schur complement $\Sigma/\Sigma_N = \Sigma_R - C \Sigma_N^{-1} C^\top$ is unique given (Σ_R, C, Σ_N) .

Proof. Congruence by $\begin{pmatrix} I & 0 \\ -C^\top \Sigma_R^{-1} & I \end{pmatrix}$ block-diagonalizes Σ to $\Sigma_R \oplus (\Sigma_N - C^\top \Sigma_R^{-1} C)$; positivity of a block matrix is equivalent to positivity of both diagonal blocks [4, 5]. Convexity of the feasible cone is immediate from the affine constraint. Uniqueness of the completion from its blocks is definitional. $\square$

Theorem 3.3 (Classification, iff). *Two resolution geometries are gauge-equivalent (related by $(U, V) \in \text{Aut}(\mathcal{G})$) if and only if they share the spectrum of Σ_R , the spectrum of Σ_N with multiplicities, the generalized canonical correlations, and the discrete gauge. This holds on every stratum, generic and non-generic.*

Proof. On the generic stratum, the singular value decomposition of the whitened coupling $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$ gives the canonical correlations as a complete invariant [7]. On a stratum with an eigenvalue of multiplicity m , the stabilizer enlarges to $O(m)$; by the Luna slice theorem [6] the neighborhood is the associated bundle with slice $\text{Sym}_0(m)$, and the coupling block is classified up to the residual $O(m)$ by its singular values (a matrix modulo one-sided orthogonal action is determined by, and determines, its Gram matrix). Both implications hold. $\square$

4 Second-order layer: the fundamental theorem of submanifolds

The second-order residue is the pair $(\mathbb{I}, \nabla^\perp)$: the second fundamental form $\mathbb{I}(X, Y) = (\nabla_X dY(\cdot))^N$, valued in the null bundle, and the normal connection $\nabla^\perp$. The integrability equations are, in the standard form for an isometric immersion of arbitrary codimension,

$$\text{(Gauss)} \quad \langle R(X, Y)Z, W \rangle = \langle \mathbb{I}(X, W), \mathbb{I}(Y, Z) \rangle - \langle \mathbb{I}(X, Z), \mathbb{I}(Y, W) \rangle, \quad (1)$$

$$\text{(Codazzi)} \quad (\bar{\nabla}_X \mathbb{I})(Y, Z) = (\bar{\nabla}_Y \mathbb{I})(X, Z), \quad (2)$$

$$\text{(Ricci)} \quad \langle R^\perp(X, Y)\xi, \eta \rangle = \langle [A_\xi, A_\eta]X, Y \rangle, \quad (3)$$

where R is the curvature of the induced (Fisher) metric, $R^\perp$ the normal curvature, A_ξ the shape operator, and $\bar{\nabla}$ the connection on $T^* \otimes T^* \otimes N$.

Theorem 4.1 (Fundamental theorem of submanifolds, iff). *Let M be a simply-connected Riemannian manifold with metric g , a vector bundle N over M with a metric connection $\nabla^\perp$, and a symmetric N -valued bilinear form $\mathbb{I}$. There exists an isometric immersion of M into the model space with normal bundle N , second fundamental form $\mathbb{I}$, and normal connection $\nabla^\perp$ if and only if $(g, \mathbb{I}, \nabla^\perp)$ satisfy the Gauss, Codazzi, and Ricci equations. The immersion is unique up to an ambient isometry.*

Proof. This is the classical fundamental theorem of submanifolds (Bonnet in codimension one; the general codimension is due to the integrability of the associated Iso-valued connection form) [1, 2, 3]. Existence is the integration of a flat Iso-connection built from $(g, \mathbb{I}, \nabla^\perp)$, whose flatness is exactly Gauss–Codazzi–Ricci; uniqueness is the uniqueness of that integration up to the constant of integration, an ambient isometry. $\square$

5 The fundamental theorem of resolution geometry

Theorem 5.1 (Synthesis as an equivalence). *Fix $\dim \Theta, \dim O$ and a simply-connected parameter domain, and adopt Assumption 2.1. Then data*

$$(\Sigma_R \succ 0, C, \Sigma_N; \mathbb{I}, \nabla^\perp)$$

is realized by a resolution geometry if and only if

- (i) (first order) $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$ and $R \subseteq \text{range } \Sigma$; and
- (ii) (second order) $(g, \mathbb{I}, \nabla^\perp)$ satisfy Gauss–Codazzi–Ricci, where $g = \Sigma^{-1}$ is the Mahalanobis metric and the intrinsic curvature R in the Gauss equation is that of the induced Fisher metric of Lemma 2.2.

When it exists, the geometry is unique up to gauge $\text{Aut}(\mathcal{G})$ and ambient isometry.

Proof. Sufficiency. Given (i), Theorem 3.2 yields a positive (semi)definite Σ with the prescribed blocks, hence by Assumption 2.1 a metric $g = \Sigma^{-1}$ and, by Lemma 2.2, an induced Fisher metric on Θ . Given (ii), Theorem 4.1 yields an isometric immersion Y realizing $(g, \mathbb{I}, \nabla^\perp)$ on the simply-connected domain. The immersion supplies $R = \text{Im } dY$ and $N = R^{\perp_g}$, and Σ supplies the covariance; together they are a resolution geometry realizing the data. *Necessity.* For an actual resolution geometry, Σ is a genuine covariance, so (i) holds by Theorem 3.2; and Y is an actual immersion, so its $(\mathbb{I}, \nabla^\perp)$ satisfy Gauss–Codazzi–Ricci identically, giving (ii). *Uniqueness.* By Theorem 3.2 the covariance is determined by its blocks; by Theorem 4.1 the immersion is determined up to ambient isometry; by Theorem 3.3 the remaining freedom is exactly the gauge $\text{Aut}(\mathcal{G})$. $\square$

Remark 5.2 (This is an assembly). The novelty is not a new deep theorem but the assembly: the Mahalanobis identification aligns the two geometries (Lemma 2.2) so that the Schur equivalence and the submanifold equivalence apply to the same object, and Theorem 3.3 accounts for the residual freedom. The credibility comes precisely from the fact that each engine is a pre-existing equivalence.

6 Consistency: are the two conditions compatible?

A derivation that combines two constraint systems must show they do not conflict.

Proposition 6.1 (No over-determination). *Conditions (i) and (ii) of Theorem 5.1 constrain disjoint data—(i) the covariance block Σ_N given (Σ_R, C) , (ii) the forms $(\mathbb{I}, \nabla^\perp)$ given g —and are linked only through the Gauss equation, which ties $\mathbb{I}$ to the curvature of $g = \Sigma^{-1}$. This link is a solvable relation, not a conflict: for any admissible Σ , the choice $\mathbb{I} = 0$ (a totally geodesic immersion) satisfies Gauss–Codazzi–Ricci trivially, so the second-order condition is always satisfiable given the first. Hence the system is consistent; no contradiction arises.*

Proof. $\mathbb{I} = 0$ makes the right-hand sides of Gauss and Ricci vanish and Codazzi trivial, which is consistent iff the induced metric is flat; for curved induced metric the Gauss equation requires $\mathbb{I} \neq 0$, and a solution $\mathbb{I}$ exists because the model space has enough codimension (the null sector) to realize any symmetric form compatible with the prescribed intrinsic curvature—the existence half of Theorem 4.1. Thus feasibility of (ii) is implied by, and never excluded by, (i). $\square$

Remark 6.2 (The only degeneracy). The single place the construction breaks is the support boundary $R \not\subseteq \text{range } \Sigma$, where Σ^{-1} and hence the Mahalanobis metric fail to exist. This is not a contradiction of the theorem but the boundary of its hypothesis (Assumption 2.1); it coincides with the finite-information boundary and the irreducible-core locus, which are external by prior results.

7 Globalization without hand-waving

The atlas was previously supported by numerical cocycle checks and topological language. Theorem 4.1 replaces both with a derivation.

Theorem 7.1 (The atlas is a genuine cocycle by uniqueness). *Cover the (not necessarily simply-connected) parameter base by simply-connected patches. On each patch Theorem 4.1 gives an immersion, unique up to ambient isometry. On a double overlap the two immersions realize the same $(g, \mathbb{I}, \nabla^\perp)$, hence differ by a unique ambient isometry $g_{\alpha\beta} \in \text{Iso}$; on a triple overlap uniqueness forces $g_{\alpha\beta}g_{\beta\gamma}g_{\gamma\alpha} = \text{id}$. The transition data $\{g_{\alpha\beta}\}$ is therefore a genuine $\text{Aut}(\mathcal{G})$ -valued Čech cocycle—by the theorem, not by numerical test.*

Proof. Two immersions with identical fundamental forms differ by an ambient isometry (uniqueness in Theorem 4.1); the isometry is unique because the immersions are full. The triple-overlap identity is the composition of three such uniqueness statements around a common region. $\square$

Corollary 7.2 (The character variety is the monodromy obstruction). *A global immersion over the base exists iff the cocycle $\{g_{\alpha\beta}\}$ is trivial, i.e. iff the monodromy representation $\rho : \pi_1(\text{base}) \rightarrow \text{Aut}(\mathcal{G})$ is trivial. The moduli of such geometries is $\text{Hom}(\pi_1, \text{Aut}(\mathcal{G}))/\text{conjugation}$ —the character variety. It is thus not an ad hoc invariant but the standard monodromy obstruction to integrating the flat Iso-connection of Theorem 4.1 over a non-simply-connected base.*

8 The precise necessity–sufficiency verdict

Theorem 8.1 (Local iff; global iff modulo monodromy). *On a simply-connected parameter domain the synthesis is a genuine equivalence: data satisfying (i)–(ii) of Theorem 5.1 is realized by a resolution geometry, uniquely up to gauge, and every resolution geometry yields such data. Over a non-simply-connected base the same equivalence holds modulo the monodromy obstruction of Corollary 7.2: local realizability and uniqueness are unchanged, and the only additional datum is the conjugacy class of $\rho : \pi_1 \rightarrow \text{Aut}(\mathcal{G})$.*

Proof. Immediate from Theorems 5.1 and 7.1 and Corollary 7.2. $\square$

This answers the question that motivated the paper. The synthesis is not merely one-way: it is a full equivalence locally, and globally it is an equivalence up to a single, precisely identified obstruction.

9 Scope and the one open computation

Non-claim 9.1. Every implication above is a proof or an assembly of established equivalences (Schur completion; the fundamental theorem of submanifolds; the Luna slice theorem), with no reliance on numerical verification. Three honest limits remain. First, Assumption 2.1 requires $\Sigma \succ 0$; the support boundary is outside the hypothesis and coincides with the known finite-information / irreducible-core locus. Second, the global equivalence holds modulo monodromy; the *existence and role* of the character variety are now derived (Corollary 7.2), but its *computation* for a given base is the classical, still-open problem of computing $\text{Hom}(\pi_1, \text{Aut}(\mathcal{G}))/\text{conj}$ —a problem of mathematics at large, not of this derivation. Third, no physical claim is made; the fluctuation–dissipation identity of the companion paper is the only physical content in the program, and it is not invoked here.

10 Conclusion

The spine can be proved, not merely illustrated. Assembling the Schur completion equivalence, the fundamental theorem of submanifolds, and the Luna slice theorem, and aligning the statistical and embedding geometries by the Mahalanobis metric, gives a fundamental theorem of resolution geometry: the first- and second-order residues realize a geometry, uniquely up to gauge, exactly when they satisfy the Schur floor and the Gauss–Codazzi–Ricci equations.

The two conditions are consistent, linked by Gauss and not over-determined. Globalization is then a consequence, not an addition: local immersions are unique, so their overlaps form a genuine $\text{Aut}(\mathcal{G})$ -cocycle, and the character variety is the monodromy obstruction to integrating the associated flat connection. The synthesis is therefore a genuine equivalence—locally exact, globally exact modulo a single identified obstruction—and the one thing that remains open, the computation of that obstruction, is openly the hard problem it has always been.

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